FA-1 SLIP TEST

Class: 8 | Subject: Mathematics
Time: 45 Minutes | Total Marks: 20 Marks

Name of the Student: .....
Roll Number: .....
Date: .....

Problem 1 [4 Marks]

Find four rational numbers lying between $^{1}/_{4}$ and $^{1}/_{2}$.

Problem 2 [4 Marks]

Using suitable properties, evaluate:

$$(-^{2}/_{3}) \times {}^{3}/_{5} + {}^{5}/_{2} - {}^{3}/_{5} \times {}^{1}/_{6}$$

Problem 3 [4 Marks]

Solve for x:

$$^{(2x-3)}/_{5} + {}^{(x+3)}/_{4} = {}^{9}/_{10}$$

Problem 4 [4 Marks]

Solve for y:

$$^{(3y+4)}/_{(2-6y)} = -^{2}/_{5}$$

Problem 5 [4 Marks]

The perimeter of a rectangle is **154 m**. Its length is **2 m more than twice its breadth**. Find the length and breadth.

FA-1 SLIP TEST — SOLUTIONS

Marks: 5 × 4 = 20 Marks

Solution 1:

Convert fractions to equivalent rational numbers with common denominator 40.

$^{1}/_{4} = {}^{(1 \times 10)}/_{(4 \times 10)} = {}^{10}/_{40}$

$^{1}/_{2} = {}^{(1 \times 20)}/_{(2 \times 20)} = {}^{20}/_{40}$

Choose four integers between 10 and 20 for numerators: 11, 12, 13, and 14.

Four rational numbers: $^{11}/_{40}$, $^{12}/_{40}$, $^{13}/_{40}$, $^{14}/_{40}$

Solution 2:

Rearrange using Commutative Property:

$(-^{2}/_{3}) \times {}^{3}/_{5} - {}^{3}/_{5} \times {}^{1}/_{6} + {}^{5}/_{2}$

Take common factor $^{3}/_{5}$ using Distributive Property:

$^{3}/_{5} \times [(-^{2}/_{3}) - {}^{1}/_{6}] + {}^{5}/_{2}$

$= {}^{3}/_{5} \times [{}^{(-4 - 1)}/_{6}] + {}^{5}/_{2} = {}^{3}/_{5} \times (-{}^{5}/_{6}) + {}^{5}/_{2}$

$= -{}^{1}/_{2} + {}^{5}/_{2} = {}^{(-1 + 5)}/_{2} = {}^{4}/_{2} = 2$

Answer: 2

Solution 3:

LCM of denominators 5, 4, and 10 is 20.

Multiply entire equation by 20:

$4(2x - 3) + 5(x + 3) = 2(9)$

Expand brackets:

$8x - 12 + 5x + 15 = 18$

$13x + 3 = 18$

$13x = 18 - 3 = 15$

$x = {}^{15}/_{13}$

Solution 4:

Apply cross-multiplication:

$5 \times (3y + 4) = -2 \times (2 - 6y)$

Expand both sides:

$15y + 20 = -4 + 12y$

Transpose variable terms to LHS and constants to RHS:

$15y - 12y = -4 - 20$

$3y = -24$

$y = -{}^{24}/_{3}$

$y = -8$

Solution 5:

Let breadth = b meters.